

- Q.28 Describe various factors affecting frequency of irrigation.
- Q.29 Describe the volumetric method of water measurement.
- Q.30 Describe Border method of irrigation alongwith its adaptability.
- Q.31 What are the different characteristics of aquifer influencing yield of well?
- Q.32 State the effects of excessive use of fertilizers on environment.
- Q.33 Define Infiltration rate and Evapotranspiration.
- Q.34 What are different methods of improving duty ?
- Q.35 Give classification of pumps.

SECTION-D

Note: Long answer type questions. Attempt any two out of three questions. (2x10=20)

- Q.36 Describe the basic details, characteristics, Layout and adaptability of sprinkler irrigation system.
- Q.37 Explain the principle of operation, working and constructional details of a centrifugal pump.
- Q.38 List different surface methods of irrigation. Explain the basic details, characteristics, type and adaptability of furrow irrigation method.

No. of Printed Pages : 4
Roll No.

180132/120132/
030132/030154

3rd Sem / Agri.

**Subject : Farm Irrigation Engineering / Water
Res . Engg.**

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice Questions. All questions are compulsory (10x1=10)

- Q.1 The ratio of the quality of water delivered to the field and quantity of water pumped into the canal.
- Water storage efficiency
 - Water application efficiency
 - Water conveyance efficiency
 - Water use efficiency
- Q.2 The irrigation method that resembles like rain.
- Sprinkler
 - Drip
 - Shallow
 - Rotary
- Q.3 Which is the method to improve the duty of water ?
- Lining of canal
 - Construction of weirs
 - Construction of dams
 - Check dams

(00) (4) 180132/120132/
030132/030154

(1) 180132/120132/
030132/030154

- Q.4 The line connecting points having the same depth below a water surface.
 a) Isobar b) Isobath
 c) Contour d) Drawdown
- Q.5 The height at which water stand in a piezometer
 a) Static water level
 b) Piezometric surface
 c) Pumping water level
 d) None of these
- Q.6 The downward movement of water is called.
 a) Seepage b) Duty
 c) Delta d) Infiltration
- Q.7 The total depth of water application to the crop is called.
 a) Duty b) Delta
 c) Base d) None of these
- Q.8 The difference between static water level and pumping water level.
 a) Area of influence b) Drawdown
 c) Well yield d) None of the above
- Q.9 Which irrigation method uses supply ditch, borders, ridges?
 a) Border irrigation
 b) Check irrigation
 c) Basin irrigation
 d) Drip irrigation
- Q.10 The Sprinkler irrigation method is recommended in
 a) Highly undulating area
 b) Soils having high infiltration rate
 c) Irrigation source is a tubewell
 d) In all the above situation

(2)

180132/120132/
030132/030154

SECTION-B

Note: Objective type questions. All questions are compulsory. (10x1=10)

- Q.11 Define soil permeability.
- Q.12 State two demerits of over irrigation.
- Q.13 List two factors affecting frequency of irrigation.
- Q.14 Write down two objectives of canal lining.
- Q.15 List different sources of irrigation water.
- Q.16 What is the use of priming?
- Q.17 Give the names of different types of impellers.
- Q.18 State two benefits of sprinkler irrigation.
- Q.19 Give two types of canal lining
- Q.20 State two quality parameter of irrigation water.

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Describe the constructional detail of diffusor type centrifugal pump.
- Q.22 Write down the advantages of canal lining.
- Q.23 What is the criteria for selection of irrigation pumps?
- Q.24 What is the necessity of irrigation ?
- Q.25 State the limitation of drip irrigation system.
- Q.26 Describe the working of submersible pumps.
- Q.27 Write different method of groundwater recharge.

(3)

180132/120132/
030132/030154